

Highlights

Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study

- Recursive degradation exhibits a sharp pressure-dependent transition
- Pressure thresholds differ by an order of magnitude across three backbone families
- Adapter rank and learning rate interact to determine the operating regime
- Output drift accompanies degradative regimes beyond factual retention loss
- Reducing synthetic exposure by 5% restores near-homeostatic behavior on Qwen

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ABSTRACT

Recursive fine-tuning on synthetic data enables scalable model adaptation. However, recursively retraining models on their own outputs can progressively degrade factual knowledge. The conditions under which recursive degradation emerges, remains bounded, or can be reversed under parameter-efficient fine-tuning remain poorly understood. Through systematic dose-response experiments spanning three backbones (Qwen 2.5 1.5B, Gemma 3 1B, and Gemma 4 E2B), ten recursive generations, and three to five independent seeds per condition, we investigate how recursive factual retention changes as adapter capacity, perturbation magnitude, and synthetic exposure vary. These experiments reveal a common empirical organizing principle, which we term effective training pressure. We find: (1) a sharp regime transition from homeostatic retention (above 90% over ten generations) to progressive degradation as adapter rank increases, with the threshold differing by roughly an order of magnitude across architectures (effective rank 3–6 on Gemma versus 50–88 on Qwen); (2) full fine-tuning learning-rate sweeps on both backbones that reproduce the same qualitative pattern, confirming that the transition is not unique to low-rank adaptation; (3) a rank $\times$ learning-rate interaction demonstrating that update capacity and perturbation magnitude jointly determine the operating regime rather than acting as independent axes; (4) a retention/distribution dissociation at intermediate pressure, where output efficiency degrades threefold while factual retention remains near 90%; and (5) restoration of near-homeostatic behavior through approximately 5% reduction in synthetic exposure at the Qwen boundary configuration. These findings characterize recursive degradation as a pressure-dependent phenomenon whose stability depends on a multidimensional control landscape, providing an empirical basis for monitoring and configuring recursive training pipelines.

1. Introduction

Large language models are frequently adapted using synthetic data generated by language models themselves. Iterative reinforcement learning from human feedback [1], synthetic instruction generation [2, 3], self-play, and model distillation all create recursive fine-tuning workflows in which successive model generations are trained on outputs produced by their predecessors. As the availability of high-quality human-generated text becomes increasingly constrained [4], these recursive workflows are becoming a common component of model adaptation pipelines.

In practice, recursive synthetic adaptation is attractive because it is inexpensive, scalable, and avoids the continuous collection of human annotations [5]. However, every recursive update also modifies the distribution from which future training data will be drawn, creating a feedback loop in which small errors may accumulate over successive generations. For practitioners, the resulting engineering question is not simply whether recursive training degrades models, but under what conditions such degradation becomes operationally relevant.

Recent work has established that this risk is both empirically observable and theoretically grounded. Under full fine-tuning, models progressively lose distributional diversity [6] and factual accuracy [7], with

expected loss growing linearly across recursive generations [8]. Similar dynamics have been observed in diffusion models and GANs [9], suggesting that recursive degradation is not unique to language models. Mitigation strategies based on data accumulation [10], verification [11], and loss shaping [12] have been proposed, each operating on a different axis of the training pipeline. Together, these studies confirm that recursive degradation is a real and well-characterized phenomenon.

Despite these findings, modern adaptation pipelines rely predominantly on parameter-efficient methods such as LoRA [13] and QLoRA [14], which dramatically reduce computational cost while preserving downstream performance. Biderman et al. [15] observed qualitatively that LoRA “learns less and forgets less,” but it remains unknown how this protective property scales under repeated recursive pressure. At what adapter capacity does the protection fail? Is the boundary between stability and degradation gradual or sharp? Is it universal across architectures, or backbone-dependent? To our knowledge, no systematic dose-response analysis has characterized the sharpness, backbone-dependence, or reversibility of the stability boundary under parameter-efficient fine-tuning.

Rather than treating adapter rank, learning rate, and synthetic exposure as independent factors, we show that their effects on recursive stability can be understood through a single organizing principle, which

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we term *effective training pressure*. Through systematic experiments spanning three backbones (Qwen 2.5 1.5B, Gemma 3 1B, Gemma 4 E2B), a strict replace-only protocol over ten recursive generations, and multiple intervention strategies, we identify abrupt backbone-dependent transitions between homeostatic and degradative recursive regimes and show that these transitions can be controlled through multiple independently manipulated variables.

Beyond explaining the observed behavior, this perspective has practical implications for the control of recursive training systems. Existing pipelines offer little guidance for selecting adapter capacity or synthetic-data reuse policies that preserve long-term stability. By framing recursive stability as a function of effective training pressure, these findings provide a basis for monitoring, controlling, and intervening in recursive fine-tuning before substantial degradation occurs. Critically, these results suggest that recursive training stability can be influenced through configuration choices already available to practitioners, without requiring external verifiers, additional real data, or changes to the training objective, provided that the effective training pressure threshold for the target backbone has first been identified.

The main contributions of this work are:

1. We provide a systematic dose-response mapping between adapter rank and recursive factual retention, measured via a subset-based retention metric (K_0 , Section 3.2), identifying a sharp regime transition on Qwen and replicating the same qualitative structure on Gemma 3 at a substantially lower threshold (effective rank 3–6 versus 50–88), confirmed across three to five independent seeds per condition.
2. We demonstrate that full fine-tuning learning-rate sweeps on both Qwen and Gemma 3 reproduce the same dose-response pattern, confirming that the phenomenon responds to perturbation magnitude rather than being specific to low-rank adaptation.
3. We show through a rank $\dot{I}$ learning-rate interaction experiment that update capacity and perturbation magnitude jointly determine the operating regime, establishing that the pressure components are not independent.
4. We show that output-distribution diagnostics reveal a retention/distribution dissociation at intermediate capacity, where output quality degrades substantially while factual retention remains bounded.
5. We show that the regime boundary is sensitive to marginal changes in synthetic exposure: a reduction of approximately 5% restores near-homeostatic retention at the Qwen boundary, although this sensitivity does not transfer across backbones without recalibration.

The remainder of this paper reviews related work, formalizes the proposed recursive fine-tuning framework and the effective training pressure formulation, reports the empirical evaluation across recursive regimes, backbone architectures, and intervention strategies, and concludes by discussing the practical implications and limitations of the proposed framework.

2. Related Work

A growing body of work has investigated the effects of recursively training language models on synthetic data [4], together with strategies for mitigating the resulting degradation. This section reviews the literature most relevant to the present study, covering recursive model collapse, factual knowledge degradation, mitigation approaches, and parameter-efficient fine-tuning, before positioning the contribution of this work within that landscape.

2.1. Recursive model collapse

Recent work has established recursive training on synthetic data as a distinct failure mode of generative models. Empirically, Shumailov et al. [6] demonstrated that language models trained iteratively on their own outputs exhibit irreversible distributional collapse, with the tails of the original data distribution progressively disappearing. This phenomenon extends beyond language: Alemohammad et al. [9] observed analogous quality and diversity degradation in self-consuming loops for diffusion models and GANs, suggesting that recursive collapse is a general consequence of iterative generative training rather than a language-specific artifact [16].

Theoretical work has formalized these observations. Dohmatob et al. [8] proved that expected test loss increases at least linearly with the number of recursive generations for any nonzero fraction of synthetic data, disrupting the neural scaling laws that underpin modern LLM training [17, 18]. Seddik et al. [19] showed convergence toward a degenerate distribution under iterated retraining, while Xu et al. [20] provided a random-walk interpretation demonstrating that the probability of model improvement under recursive training is strictly less than one-half. Dey and Donoho [21] further characterized the growth schedules required to avoid collapse in exponential-family models.

Collectively, these studies establish that recursive degradation is both empirically observable and theoretically expected under full fine-tuning or pretraining. However, they provide limited insight into where the transition between stable and degradative behavior occurs under parameter-efficient fine-tuning, or how that transition depends on controllable training variables such as update capacity and perturbation magnitude.

2.2. Knowledge degradation

Whereas much of the recursive-training literature examines statistical consequences of repeated synthetic training (distributional decay, mode collapse), a smaller body of work has focused on how recursive updates affect the retention of factual knowledge. Although not concerned with recursive training, Petroni et al. [22] established that pretrained language models encode factual associations accessible via probing, providing the conceptual basis for knowledge-retention evaluations. Building on this foundation, Keisha et al. [7] introduced the term *knowledge collapse* to describe a three-stage process in which factual accuracy deteriorates while surface fluency persists. Their experiments on Gemma 3 1B under full fine-tuning identified a critical Stage B where the model retains linguistic competence but loses factual grounding. Beyond factual degradation itself, Kovač et al. [23] showed that properties of the recursive training stream (domain breadth, sample diversity) modulate how distribution shift unfolds across iterations, demonstrating that degradation trajectory is not fixed but depends on controllable data characteristics.

Our work is most closely related to Keisha et al. [7], as both studies examine factual degradation under recursive training. However, the two works address different questions. Whereas Keisha et al. [7] characterize stages of knowledge collapse under recursive full fine-tuning, we map how factual retention responds as effective training pressure varies under parameter-efficient fine-tuning. These differences are reflected in the training protocol (replace-only QLoRA versus accumulated FFT) and evaluation methodology (exact-match K_0 retention versus multi-format assessment). Although both studies include Gemma 3 1B, the different protocols preclude direct quantitative comparison.

2.3. Mitigation strategies

Recent work has shifted from characterizing recursive collapse to developing interventions that improve the stability of recursive synthetic training. These interventions operate at different stages of the training pipeline. At the verification stage, Yi et al. [11] demonstrated that external filtering of synthetic data substantially improves short-term recursive stability, while also identifying a long-term convergence induced by repeated reliance on the same verifier; Feng et al. [24] provided complementary theoretical analysis showing that verification is necessary for scaling with synthesized data. At the data-composition stage, Gerstgrasser et al. [10] showed that accumulating real data alongside synthetic generations bounds recursive error growth, preventing collapse when fresh human-generated data remains available. Bertrand et al. [25] proved stability conditions for iterative retraining when real data constitutes a sufficient proportion of each training batch, while Gerstgrasser et al. [10] argued that

accumulation renders collapse avoidable under realistic data-growth assumptions. At the training-objective stage, Zibakhsh Shabgahi et al. [12] introduced Truncated Cross-Entropy (TCE), a loss function that selectively ignores high-confidence tokens during training, enabling models to remain stable under substantially larger proportions of synthetic data.

These three approaches represent distinct mitigation axes: data quality (verification), data composition (accumulation), and training objective (loss shaping). Our work identifies a complementary control axis: the dynamics of parameter updates during fine-tuning. Rather than proposing a new data-selection, verification, or loss-modification method, we investigate how changes in update capacity, perturbation magnitude, and synthetic exposure can independently influence the transition between stable and degradative behavior under strict replace-only conditions.

2.4. Parameter-efficient fine-tuning and forgetting

Parameter-efficient fine-tuning (PEFT) constrains model updates to low-dimensional subspaces. Hu et al. [13] established this paradigm with LoRA, demonstrating that decomposing weight updates into low-rank matrices reduces trainable parameters by orders of magnitude without sacrificing downstream performance. The theoretical justification for such extreme parameter reduction relies on the low intrinsic dimensionality of pretrained models during fine-tuning [26]: pretrained representations already encode task-relevant structure in a compact subspace. Zhang et al. [27] generalized this insight through adaptive rank allocation across layers, guided by SVD-based importance scoring. Dettmers et al. [14] further optimized the practical deployment of low-rank adaptation through QLoRA’s 4-bit quantization, enabling fine-tuning of large models on consumer hardware.

Beyond computational efficiency, recent work has begun investigating how parameter-efficient adaptation influences knowledge stability. While classical continual learning literature has long studied catastrophic forgetting under sequential task updates [28, 29, 30], recent work explores stability under recursive data regimes. Biderman et al. [15] observed that LoRA implicitly regularizes adaptation, concluding that it “learns less and forgets less” than full fine-tuning. Recent preprints further suggest that higher LoRA ranks increase forgetting while improving plasticity [31, 32], a pattern consistent with the dose-response relationship we observe. Emerging frameworks attempt to operationalize this stability through structural matrix constraints [33]. In recursive synthetic training specifically, Adapala et al. [34] combined cumulative data retention with quality filtering in their Anti-Ouroboros framework. Mechanistically, this protection may relate to how transformers store factual associations: Geva et al.

[35] demonstrated that feed-forward layers function as key-value memories, and Meng et al. [36] localized specific factual associations to individual MLP layers via causal tracing. These findings suggest that low-rank constraints on specific modules may bottleneck the network’s ability to overwrite foundational facts.

Complementing Biderman et al. [15]’s observation that LoRA reduces forgetting relative to full fine-tuning, our work shows that this stability is conditional rather than universal. We empirically identify backbone-dependent pressure thresholds, determined by effective training pressure (a composite of adapter capacity, perturbation magnitude, and synthetic exposure), beyond which recursive synthetic fine-tuning transitions from bounded retention to progressive degradation.

2.5. Positioning of this work

The present study does not refute the model-collapse results of Shumailov et al. [6] or Dohmatob et al. [8]; it characterizes the pressure regime in which their predictions apply under practical PEFT configurations. It does not discover that LoRA mitigates forgetting [15]; it characterizes the boundary conditions under which this protection ceases to hold. It does not propose a general-purpose mitigation method; rather, it demonstrates experimentally that recursive stability varies systematically with effective training pressure along three axes: update capacity, perturbation magnitude, and synthetic exposure. These axes are orthogonal to the data-composition [10], synthetic-data verification [11], and loss-shaping [12] approaches in the existing mitigation literature.

Rather than asking whether recursive synthetic training inevitably collapses or whether PEFT universally prevents forgetting, this work asks where the transition between stable and degradative regimes occurs under controlled changes in effective training pressure.

3. Methodology

We study the stability of recursive synthetic fine-tuning through the lens of *effective training pressure*: the combined effect of update capacity, perturbation magnitude, and synthetic-data exposure applied at each recursive generation. We do not treat this quantity as a single fitted scalar. Instead, we use it as an organizing framework in which each component is manipulated separately: adapter rank controls update capacity, learning rate controls perturbation magnitude, and filtering or downsampling controls synthetic exposure. This design lets us ask whether transitions between homeostatic and degradative regimes are attributable to capacity, magnitude, exposure, or their interaction.

Our experiments follow a dose-response design across these three axes, tracking factual retention and output-distribution drift over up to ten recursive generations. In the main protocol, each generation is

trained only on synthetic outputs produced by the previous generation, with no real-data accumulation, replay buffer, or quality filter. This strict replace-only setting follows the recursive data-replacement paradigm studied in the model-collapse literature [6, 8, 10], while allowing us to isolate how the update mechanism changes the resulting dynamics.

The design is motivated by both practical and theoretical considerations. In practice, parameter-efficient fine-tuning via LoRA [13] and QLoRA [14] dominates production adaptation of large language models, yet behavior under recursive synthetic exposure remains poorly characterized beyond the observation that LoRA can “learn less and forget less” [15]. That work does not map when this regularization remains protective and when it fails under repeated synthetic training. Theoretically, fine-tuning updates often lie in low-dimensional subspaces [26], suggesting that adapter rank directly modulates the effective dimensionality of the perturbation applied each generation. We operationalize this via the effective rank of the composed update matrices BA , following the entropy-based definition of Roy and Vetterli [37].

Figure 1 provides an overview of the recursive experimental pipeline. Its distinguishing feature is that pressure manipulation (Stage 3) systematically varies training intensity across multiple axes rather than evaluating a single fixed configuration. Each axis produces an independent dose-response curve, and the resulting regime classification (Stage 5) constitutes the primary empirical contribution.

Throughout this section, the following defaults apply unless explicitly noted: the training corpus contains 2,000 questions, K_0 is defined per backbone, QLoRA uses 4-bit NF4 quantization with bfloat16 compute, and effective rank is reported as the mean across adapted matrices.

3.1. Recursive replace-only protocol

Our main recursion protocol is *data-recursive* rather than *weight-recursive*. At each generation, synthetic outputs from the previous generation are used to construct the next training set, but model updates are not accumulated across generations. In the QLoRA experiments, the base model remains frozen and a fresh adapter is initialized at each generation. In the full fine-tuning (FFT) experiments, the same replace-only data recursion is used, but the full model weights are updated instead of a low-rank adapter. This design separates degradation in the synthetic data stream from artifacts caused by adapter stacking, LoRA merging, or cumulative checkpoint continuation.

Let $\mathcal{Q}_{\text{train}}$ denote the set of training questions used to generate synthetic data. We denote by $\mathcal{S}_t$ the synthetic prompt-response dataset obtained by pairing $\mathcal{Q}_{\text{train}}$ with the model-generated responses at generation t . At generation 0, the base model answers

$\mathcal{Q}_{\text{train}}$ to produce S_0 . For each subsequent generation $t \in \{1, 2, \dots, T\}$, the protocol proceeds as follows:

1. A new trainable update mechanism is initialized: a fresh QLoRA adapter for PEFT experiments, or a fresh full fine-tuning run for FFT experiments.
2. The model is fine-tuned for 2 epochs on S_{t-1} , using standard causal language modeling loss over the formatted prompt-response sequences.
3. The resulting model is evaluated on a held-out factual retention set described in Section 3.2.
4. The fine-tuned model generates a new synthetic dataset S_t by answering the same training questions $\mathcal{Q}_{\text{train}}$ using stochastic decoding.
5. The trained update and optimizer state are not carried forward. Only the newly generated dataset S_t is used to train the next generation.

The synthetic training stream consists of 2,000 questions sampled from the TriviaQA rc.nocontext split [38], shuffled with fixed seed 15. A disjoint set of 200 questions is held out for factual retention evaluation. The held-out set is never used for synthetic-data generation or training. Synthetic responses are generated in batches of 8 using the model-specific chat template, with stochastic decoding (temperature 0.7, top- $p = 0.9$, maximum 30 new tokens). The resulting prompt-response sequences are stored and used as the training corpus for the next generation.

For QLoRA runs, adapters are trained with learning rate 10^{-5} , effective batch size 16 (micro-batch size 2 with 8 gradient-accumulation steps), bf16 precision, LoRA scaling $\alpha = 2r$, dropout 0.05, and maximum sequence length 256 tokens. No warmup, weight decay, or learning-rate scheduling is applied. FFT runs use the same recursive data protocol and evaluation procedure, but update the full model weights; FFT-specific optimizer and learning-rate settings are described in Section 3.4.

The protocol is intentionally replace-only: previous real or synthetic datasets are not accumulated, and no replay buffer is used. This contrasts with accumulation-based mitigation strategies [10] and with loss-level interventions such as truncated cross-entropy losses [12]. We use this strict setting to expose how recursive stability changes as a function of effective training pressure. Controlled interventions on synthetic exposure, including filtering and downsampling, are introduced separately in Section 3.7.

3.2. K_0 factual retention metric

Our primary evaluation metric is K_0 retention: the fraction of facts that the base model answers correctly at generation 0 and continues to answer correctly at generation t . This metric isolates factual *loss* from factual *absence*: models are evaluated only on facts they demonstrably possessed at the start of the recursive process.

At generation 0, the base model without any adapter is evaluated on a held-out set of 200 TriviaQA questions. Each question includes multiple valid answer aliases. A response is scored as correct if, after normalization, the predicted string contains any ground-truth alias or any ground-truth alias contains the predicted string. This bidirectional substring criterion is intended to handle short-answer variation and aliasing in TriviaQA-style factoid responses. Normalization consists of lowercasing, removing articles (*the*, *a*, *an*), stripping punctuation, converting written number words to digits, and collapsing whitespace.

The subset of held-out questions answered correctly at generation 0 defines K_0 . For example, for Qwen 2.5 1.5B, $K_0 = 78$ out of 200 held-out questions. Other backbones have their own fixed K_0 sets, determined once at generation 0 and held constant across all experimental conditions for that backbone.

At each subsequent generation t , only the corresponding K_0 items are used to compute retention:

$$\text{Retention}_t = \frac{|\{q \in K_0 : q \text{ correct at gen. } t\}|}{K_0} \quad (1)$$

Thus, a retention score of 100% means that the model preserved all factual items it answered correctly at generation 0. The metric does not measure global factual accuracy, nor does it reward acquisition of facts outside K_0 ; it is designed specifically to measure preservation of initially demonstrated knowledge under recursive training.

We additionally track item-level transitions within K_0 . A correct-to-wrong transition ($C \rightarrow W$) indicates that an initially retained fact was lost at a later generation. A wrong-to-correct transition ($W \rightarrow C$) indicates that a previously lost K_0 item became correct again in a later generation. These transition counts help distinguish monotonic forgetting, temporary instability, and bounded oscillation even when aggregate retention is similar.

Because the same normalization and alias-matching procedure is applied at every generation and to every condition within a backbone, our analysis emphasizes relative retention dynamics rather than absolute TriviaQA benchmark performance. We report K_0 retention as both a fraction and a percentage (e.g., $73/78 = 93.6\%$). For multi-seed experiments, we report per-seed values and summaries across seeds. K_0 is defined separately for each backbone and then reused unchanged across rank, learning-rate, module targeting, and intervention strategies.

3.3. Models and hardware

We conduct experiments on three small instruction-tuned transformer backbones in the 1–2B parameter range, spanning two model families. Each backbone serves a different evidential role: Qwen 2.5-1.5B is the primary model for detailed dose-response analysis,

Gemma 3 1B is used for cross-architecture threshold validation, and Gemma 4 E2B is included as an additional single-seed robustness check. Architectural meta-data are taken from the released model configurations.

- **Qwen2.5-1.5B-Instruct** [39] is the primary backbone used for the QLoRA rank sweep, full fine-tuning comparison, distributional-output analysis, and synthetic-exposure interventions. The model has 1.54B parameters, 28 transformer layers, hidden size 1536, and grouped-query attention with 12 attention heads and 2 key-value heads.
- **Gemma 3 1B IT** [40] is the secondary backbone used for testing whether the regime transition generalizes outside the Qwen family. The released model configuration specifies 26 layers, hidden size 1152, and grouped-query attention with 4 attention heads and 1 key-value head. We evaluate the boundary between homeostatic and degradative regimes with five seeds at the boundary ranks ($r = 4$ and $r = 16$).
- **Gemma 4 E2B IT**¹ is used as additional validation from a third model family. The released configuration specifies 35 layers, hidden size 1536, and grouped-query attention with 8 attention heads and 1 key-value head. We evaluate three ranks ($r = 4$, $r = 16$, $r = 64$) with a single seed over ten generations to test whether the dose-response structure generalizes beyond the Qwen and Gemma 3 families.

QLoRA experiments load the base model in 4-bit NormalFloat (NF4) quantization with double quantization and bfloat16 compute dtype, following the QLoRA protocol [14]. Full fine-tuning experiments train the model weights in bfloat16 without 4-bit weight quantization; memory usage is reduced through gradient checkpointing and an 8-bit paged optimizer rather than model quantization.

Experiments were run on a single NVIDIA RTX 4000 Ada GPU with 20 GB of VRAM. In our implementation, a single QLoRA generation on Qwen2.5-1.5B completes in approximately 3 minutes at low ranks ($r \leq 16$) and 12 minutes at $r = 256$; full fine-tuning generations require approximately 25 minutes with gradient checkpointing. A complete 10-generation QLoRA recursive run at $r = 16$ completes in under one hour. These timings are reported to document that the full protocol is reproducible on a single workstation-class GPU (see Appendix A for code and data access).

¹Available at <https://huggingface.co/google/gemma-4-e2b-it>. No formal technical report has been published at time of writing.

Table 1

Experimental axes and their corresponding pressure components.

#	Axis	Variable	Range
1	Capacity	QLoRA rank r	$\{4, 16, 32, 64, 128, 256\}$
2	Magnitude	FFT learning rate	10^{-6} – 2×10^{-5}
3	Method	QLoRA vs FFT	Matched pert., $N=3$
4	Topology	Target modules	Attention vs full-linear
5	Exposure	Training examples	100% vs 95%

3.4. Experimental axes

Our dose-response design manipulates effective training pressure along five experimental axes. Each axis targets a different component of pressure while keeping the recursive protocol, dataset, and evaluation procedure fixed. The axes are not intended to define a single scalar pressure value; rather, they isolate how different sources of pressure, i.e., update capacity, perturbation magnitude, module topology, and synthetic exposure, affect recursive stability. Table 1 summarizes the experimental axes, the manipulated variable, and the tested range.

The experimental design manipulates effective training pressure along five axes:

1. **Adapter rank (update capacity).** We vary QLoRA rank across six values ($r = 4, 16, 32, 64, 128, 256$) on Qwen, producing adapters from 545K to 34.9M trainable parameters. The headline ranks ($r = 16$ and $r = 256$) are evaluated with three seeds; intermediate ranks trace the dose-response curve with a single seed. On Gemma 3, we test $r \in \{4, 10, 12, 14, 16\}$ with five seeds at the boundary ranks ($r = 4$ and $r = 16$) and a single seed at intermediate ranks.
2. **Learning rate (perturbation magnitude).** For full fine-tuning, we sweep the learning rate across four logarithmically spaced values (10^{-6} , 5×10^{-6} , 10^{-5} , 2×10^{-5}), selected to span one order of magnitude within the practical range commonly used for LLM fine-tuning [13, 14]. This range produced per-generation weight drifts from 0.39 to 3.48 (L_2 norm), enabling perturbation-magnitude dose-response analysis. This axis tests whether perturbation magnitude drives the regime transition independently of adapter rank.
3. **Method comparison (QLoRA vs FFT).** We compare QLoRA $r = 16$ against FFT at LR = 10^{-6} , selected because it produces the closest available perturbation proxy ($BA_F \approx 0.42$ vs FFT drift ≈ 0.39). Because the LoRA Frobenius norm and the FFT weight drift measure different geometric quantities, this constitutes a comparison

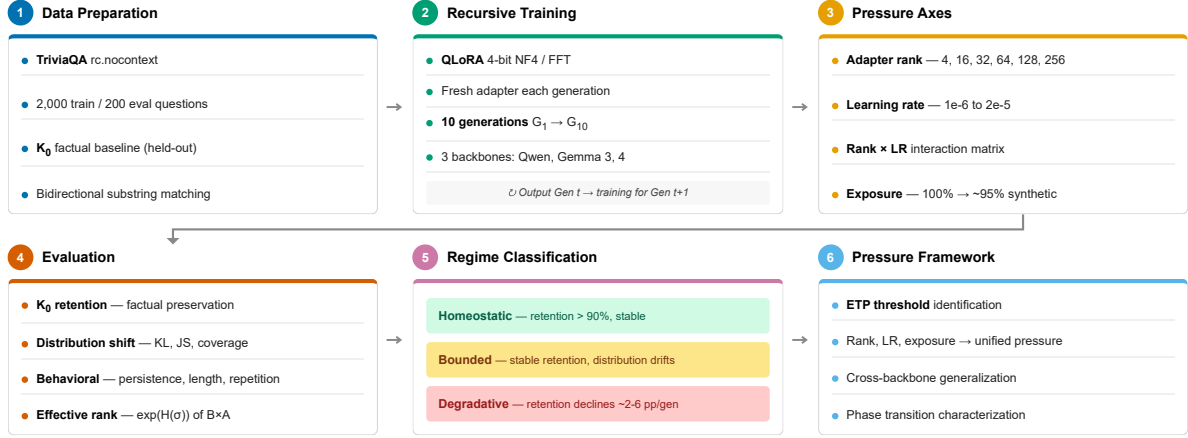


Figure 1: Overview of the experimental pipeline: (1) data preparation and K_0 baseline establishment; (2) recursive training over ten generations across three backbones; (3) pressure manipulation along five axes (rank, learning rate, method, module topology, exposure); (4) evaluation of retention, distributional shift, and effective rank; (5) regime classification; (6) effective training pressure framework synthesis.

at comparable perturbation levels rather than an exact match. Both are run for 10 generations with three seeds.

4. **Module topology (supportive ablation).** On Qwen, we compare attention-only modules (q_proj , v_proj) with all linear modules. This tests whether stability depends on module location or effective update capacity. Single-seed; supportive evidence only.
5. **Synthetic exposure (intervention).** We manipulate the synthetic training stream in the Qwen $r = 256$ boundary regime. Interventions include length filtering, random downsampling (three independent masks, i5~ removal), and short-answer constraining. Design detailed in Section 3.7.

3.5. Effective training pressure

We use *effective training pressure* as a conceptual framework for interpreting recursive fine-tuning stability, rather than as a single fitted equation. The term refers to the effective perturbation imposed on the model and the synthetic training stream at each recursive generation. In our experiments, this pressure arises from three jointly contributing factors:

1. **Update capacity:** the dimensionality of the trainable update subspace, controlled by QLoRA rank and measured through effective rank.
2. **Perturbation magnitude:** the size of the parameter update induced by one generation of training, controlled by learning rate in FFT and approximated by the norm of the composed LoRA update in QLoRA.
3. **Synthetic exposure:** the amount and composition of model-generated training data fed back into the next generation, manipulated through filtering or downsampling of the synthetic stream.

We do not assume a known functional form combining these components. Instead, our experiments manipulate each component separately and show that changes along any of the three axes can move the system between bounded and degradative regimes.

3.5.1. Effective rank

To quantify update capacity beyond nominal rank, we compute the effective rank of the composed LoRA update matrices BA after training at each generation. Following Roy and Vetterli [37], the effective rank is defined as

$$\text{erank}.BA/ = \exp\left(\sum_i \sigma_i \log \sigma_i\right), \quad (2)$$

where $\sigma_i = \sigma_i' \sigma_j$ are the normalized singular values of BA . This measure captures how many singular directions carry substantial weight in the adapter update. A rank- r adapter with uniformly distributed singular values reaches $\text{erank} = r$, whereas an update dominated by a single singular direction has $\text{erank} \approx 1$ regardless of nominal rank.

We report the mean effective rank across adapted matrices in a model. For comparisons across module-targeting schemes, we also consider aggregate effective rank, defined as the sum of effective ranks across adapted matrices. In our within-backbone rank sweeps, effective rank generally increases with nominal rank and organizes the observed transition between stable and degradative regimes better than nominal rank alone. However, raw effective-rank thresholds are not portable across architectures: Qwen remains bounded at substantially higher effective ranks than Gemma 3. We therefore treat effective rank as a within-backbone capacity measure and discuss cross-architecture normalization as a limitation in Section 6.

3.5.2. Perturbation magnitude

For full fine-tuning, we measure perturbation magnitude as the mean absolute parameter change relative to the original model weights:

$$\text{drift}.t/ = \frac{1}{\theta_i} \theta_i^{t/} * \theta_i^{0/}. \quad (3)$$

Here $\theta^{0/}$ denotes the original model parameters and $\theta^{t/}$ the parameters after one generation of fine-tuning. This scalar summarizes the average displacement of the model in weight space under FFT. For QLoRA, the closest available perturbation proxy is the Frobenius norm of the composed adapter update, BA_F . These two quantities are not geometrically identical: one measures displacement of the full model weights, while the other measures the magnitude of a low-rank update. We therefore use them only as approximate perturbation proxies when selecting the FFT/QLoRA comparison in Axis 3.

3.6. Output-distribution diagnostics

Beyond factual retention, we track output-distribution diagnostics designed to measure how the synthetic training stream changes across recursive generations. These metrics serve two purposes. First, they detect distributional degradation that may not be captured by K0 retention alone. Second, they provide monitorable signals that could be used to audit recursive fine-tuning pipelines. We treat these diagnostics as descriptive indicators of output drift, not as standalone causal explanations for factual loss. We compute four complementary diagnostics.

Content efficiency is defined as factual retention normalized by mean response length:

$$\text{ContentEff}.t/ = \frac{\text{Retention}.t/}{\text{MeanLength}.t/}. \quad (4)$$

Here, $\text{MeanLength}.t/$ is measured in words over the generated synthetic responses at generation t . Declining content efficiency indicates that the model produces more tokens per retained fact; stable content efficiency indicates that retention and response length remain proportionally stable.

Baseline response persistence compares the model's normalized response at generation t with its response at generation 0:

$$\text{Persist}.t/ = \frac{\hat{q} \ddot{E} K_0 : a_t.q/ = a_0.q/}{K_0}. \quad (5)$$

High persistence indicates responses remain anchored to generation-0 outputs. Declining persistence indicates response drift even when aggregate retention remains bounded.

Lexical diversity and verbosity are measured using several corpus-level metrics, motivated by prior characterizations of neural text degeneration [41]. Distinct- n [42] computes the ratio of unique n -grams to total

n -grams; we report Distinct-1 and Distinct-2. Because Distinct- n is sensitive to response length, we also compute MTLTD [43], a length-robust diversity estimate. We additionally track mean response length and stopword ratio to distinguish verbosity increase from content-word diversity changes.

Finally, we define the *Synthetic Drift Index* (SDI-3) to summarize output-distribution change:

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I, \quad (6)$$

where $\Delta I = \text{Instability}.T/*\text{Instability}.1/$ and $\text{Instability}.t/$ denotes the fraction of training items whose normalized response differs between generation $t*1$ and generation t . The three terms correspond to verbosity increase, unigram diversity decrease, and loss of item-level response stability. SDI-3 is not proposed as a theoretical measure; it is an empirical summary statistic used to compare output drift across experimental conditions.

These diagnostics complement K0 retention by measuring distributional properties of the synthetic stream. K0 retention measures preservation of initially known facts, whereas the output diagnostics measure whether the generated training data remain short, lexically efficient, and anchored to the model's initial responses. Throughout the paper, we interpret these metrics as signatures of distributional drift. Causal claims are reserved for explicit interventions on the synthetic training stream in Section 3.7.

3.7. Synthetic-exposure interventions

The interventions in Axis 5 test whether the degradative regime observed in the high-capacity Qwen $r = 256$ condition can be shifted back toward homeostasis by modifying the synthetic training stream. All interventions use the same backbone, rank, dataset, and recursive protocol as the Qwen $r = 256$ baseline. The initial intervention comparison is conducted with seed 15 for 5 generations; the strongest intervention is then replicated across additional seeds or random masks, as described below.

We define five intervention strategies to evaluate whether reducing synthetic exposure can restore homeostatic behavior:

- **C1 (Normal):** the unmodified baseline. Synthetic responses are generated and used for training without filtering, downsampling, or post-processing. This is the same replace-only protocol used in the rank sweep.
- **C2 (Short-answer constrained):** the generation prompt is modified to request a brief factual answer, and the maximum generation length is reduced to 10 tokens. This constrains output verbosity at generation time while retaining all training examples.

- **C3 (Length-filtered)**: synthetic responses are generated with the normal prompt, then examples whose responses exceed 5 words are removed from the next-generation training set. The remaining examples are used without duplication.
- **C4 (Canonical extraction)**: synthetic responses are generated normally, then a heuristic extracts a short factoid span from each response and replaces the original response with this canonicalized answer.
- **C5 (C3-matched random downsampling)**: synthetic responses are generated normally, then a random subset of examples is selected so that the resulting training stream approximately matches the token budget of C3. The subset is chosen by a random permutation seeded independently of the training seed.

The key comparison is between C3 and C5. C3 reduces exposure by removing long responses, whereas C5 reduces exposure by removing a random subset of examples until the training stream matches C3's approximate token budget. Thus, if both conditions produce similar stabilization, the effect cannot be attributed solely to removing long responses; it instead suggests sensitivity to marginal synthetic exposure in this boundary regime.

To test robustness, C3 is evaluated with three independent training seeds. C5 is evaluated with three independent random downsampling masks. We compute Jaccard overlap between the examples removed by C3 and by each C5 mask to verify that the interventions remove largely distinct subsets of the training stream.

C2 and C4 provide additional controls on output format. C2 changes the generation instruction and length limit while retaining all examples; C4 canonicalizes responses after generation. These conditions help distinguish exposure reduction from response-format control. We report token counts, example counts, and retention for all intervention strategies to assess whether improvements can be explained by reduced training volume alone.

3.8. Seeds, statistics, and reporting

Recursive fine-tuning experiments are computationally expensive because each condition requires multiple generations of training and data regeneration. We therefore use a small but explicit seed policy, distinguishing between multi-seed headline comparisons and single-seed supporting ablations.

All headline comparisons are evaluated with multiple independent training seeds (three for Qwen conditions, five for Gemma 3 boundary ranks). These include the Qwen rank-sweep endpoints ($r = 16$ and $r = 256$), the Gemma 3 boundary ranks ($r = 4$ and $r = 16$), the QLoRA-FFT comparison, and the C3/C5

synthetic exposure interventions. Intermediate dose-response points and supporting ablations are evaluated with a single seed and labeled as such. We do not pool single-seed and multi-seed results when making headline claims.

For multi-seed conditions, we report per-seed values together with the mean and observed range. When the observed ranges of two conditions do not overlap across all seeds, we interpret this as evidence of consistent empirical separation under the present experimental protocol.

We emphasize effect sizes, ordinal dose-response behavior, and per-seed consistency. The QLoRA rank sweep and FFT learning-rate sweep are interpreted as dose-response curves across six and four levels respectively, where the ordinal consistency of the entire sweep provides stronger evidence than any single pairwise comparison.

Effect sizes are reported as absolute percentage-point gaps in factual retention. For QLoRA-FFT comparisons, gaps are reported per seed and as a mean. For synthetic-exposure interventions, we report the improvement over the unmodified C1 baseline at the same generation. We do not claim precise threshold locations or a universal effective-rank value at which degradation begins; instead, we identify the interval between adjacent tested settings where the regime transition occurs. Single-seed observations are used as exploratory or supportive evidence and are not used as standalone support for headline conclusions.

The results presented in Section 4 follow the experimental organization introduced above. They first examine capacity-gated regime transitions (Axes 1–2), then compare QLoRA and full fine-tuning (Axis 3), characterize the accompanying distributional signatures, and finally evaluate exposure interventions (Axis 5). Module topology ablations (Axis 4) are presented alongside the capacity analysis.

4. Results

The results are organized by experimental axis to examine how the main components of effective training pressure relate to recursive stability and degradation. We begin with the central empirical result: a clear, monotonic regime transition as adapter rank increases on Qwen, which serves as the primary backbone because it provides the highest-resolution dose-response characterization through six rank levels evaluated across ten generations (Section 4.1). We then evaluate whether the same behavior generalizes across different model families (Section 4.2), investigate whether increasing perturbation magnitude through full fine-tuning reproduces the same dose-response pattern (Section 4.3), characterize the distributional signatures associated with above-threshold regimes (Section 4.4), and present evidence that a marginal reduction in

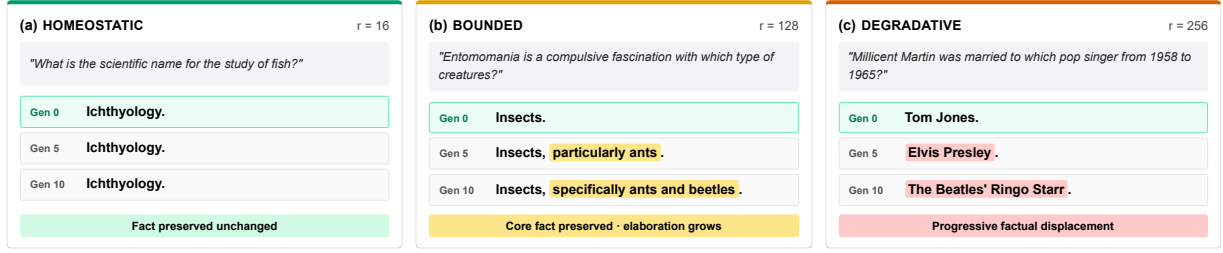


Figure 2: Representative response evolution under the three recursive regimes (Qwen 2.5 1.5B, seed 15). Examples illustrate the characteristic behavior of each regime: stable retention (a), distributional drift with bounded accuracy (b), and progressive factual displacement (c). Examples were selected from the K_0 evaluation benchmark to illustrate the median behavior within each regime classification.

synthetic exposure restores near-homeostatic behavior at the regime boundary (Section 4.5). Throughout this section, headline findings are supported by multi-seed experiments, whereas single-seed results are presented as supporting evidence.

Figure 2 illustrates how the same factual question produces qualitatively different response trajectories across regimes: in (a), the correct answer persists unchanged; in (b), the answer remains correct but becomes embedded in increasingly verbose filler text; in (c), the factual content is progressively displaced by unrelated or fabricated information.

4.1. Update capacity: rank dose-response

Increasing adapter rank produces a clear, monotonic decline in factual retention across the full rank sweep on Qwen 2.5 1.5B. Table 2 summarizes K_0 retention, effective rank, trainable parameters, and regime classification for each tested configuration. Across the evaluated range, Gen10 retention decreases from 96.2% at $r = 16$ to 79.1% at $r = 256$, a difference of 17.1 percentage points.

Table 2

QLoRA rank sweep on Qwen 2.5 1.5B-Instruct. The degradative endpoint ($r = 256$) is replicated with three seeds to confirm reproducibility.*

Rank (r)	Params	Gen	Retention	Eff. Rank	Regime
4	545K	5	97.4%	3.34	Homeo.
16	2.2M	10	96.2%	11.08	Homeo.
32	4.4M	5	96.2%	17.85	Homeo.
64	8.7M	10	92.3%	29.52	Homeo.
128	17.4M	10	89.7%	50.16	Bounded
256*	34.9M	10	79.1%	87.57	Degrad.

*Mean across three independent seeds.

The longitudinal trajectories (Fig. 3) reveal distinct temporal dynamics across regimes.

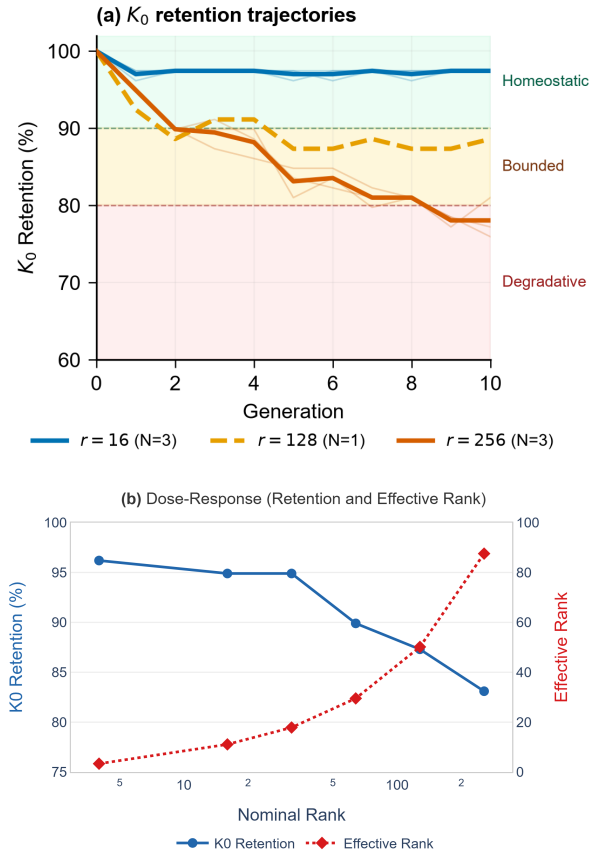


Figure 3: (a) K_0 retention trajectories over 10 recursive generations for three representative ranks on Qwen 2.5 1.5B. Semi-transparent lines show individual seeds; solid thick lines show the per-rank mean. Background bands indicate regime classification (green: homeostatic, yellow: bounded, red: degradative). (b) Dose-response relationship: Gen10 retention versus nominal rank (left axis); mean effective rank versus nominal rank (right axis). Effective rank grows sub-linearly with nominal rank, indicating that increases in nominal capacity translate into progressively smaller increases in effective update dimensionality.

At $r = 16$, retention stabilizes after an initial one-generation adaptation cost and remains flat through Gen10 (96.2%). At $r = 128$, retention settles at 88–89% by Gen5 and remains bounded through Gen10,

neither recovering nor declining further. This bounded behavior distinguishes it from both the stable regime and the progressively degradative regime. At $r = 256$, retention declines progressively at approximately 2 percentage points per generation on average, reaching 79.1% (mean across three seeds) by Gen10 with no sign of stabilization within the observed horizon.

The dose-response curve (Fig. 3b) reveals several additional features. First, the relationship between rank and retention is monotonically decreasing across all six values, spanning more than an order of magnitude in trainable parameters (545K to 34.9M). No rank reversal or non-monotonicity is observed. Second, the decline accelerates at higher ranks: the gap between $r = 64$ (92.3%) and $r = 256$ (79.1%) is substantially larger than any gap between adjacent lower-rank configurations. Third, effective rank increases sub-linearly with nominal rank (3.34 at $r = 4$ to 87.57 at $r = 256$), indicating that higher-rank adapters do not fully utilize their nominal capacity while still producing substantially larger effective updates.

The multi-seed experiments at $r = 256$ confirm that the degradative regime is reproducible. Mean Gen10 retention is 79.1% (range 76.9–82.1% across three seeds). To verify that the observed homeostatic regime was not specific to the original run, we independently repeated the $r = 16$ configuration with three random seeds as part of the QLoRA-versus-FFT experiments (Section 4.3); all three runs converged to 97.4% Gen10 retention. The ranges do not overlap: the lowest $r = 16$ value (97.4%) exceeds the highest $r = 256$ value (82.1%) by 15.3 percentage points, demonstrating clear empirical separation under the present experimental protocol.

As a supportive ablation, we compared attention-only and full-linear module targeting on Qwen at a single seed. Targeting all linear modules ($\mathbf{q}, \mathbf{k}, \mathbf{v}, \mathbf{o}, \text{gate}, \text{up}, \text{down}$) at $r = 4$ produces 4.6M trainable parameters and 96.2% Gen5 retention, matching attention-only targeting at $r = 16$ (2.2M parameters, same retention). The full-linear $r = 4$ adapter has a mean effective rank of 3.43, compared to 11.08 for attention-only $r = 16$. This preliminary observation is consistent with effective update capacity, rather than module location or parameter count alone, being the primary predictor of regime membership. Because it is based on a single seed, we treat it as supportive evidence.

Table 3 summarizes item-level transitions across the rank sweep. In the homeostatic regime, facts lost at one generation are recovered at similar rates (net loss ≈ 0), indicating bounded oscillation rather than progressive forgetting. In the degradative regime, losses consistently exceed recoveries, producing monotonic net displacement.

The rank sweep and item-level transitions establish that increasing update capacity through adapter rank is sufficient to produce a monotonic transition from

Table 3

Item-level transitions across recursive generations on Qwen 2.5 1.5B (seed 15, $K_0 = 78$). C→W (Correct→Wrong): facts lost; W→C (Wrong→Correct): facts recovered.

Rank	Gens	C→W	W→C	Net loss
$r = 4$	5	1	0	1
$r = 16$	10	3	3	0
$r = 32$	5	2	3	*1
$r = 64$	10	7	7	0
$r = 128$	10	10	7	3
$r = 256$	10	19	4	15

homeostatic to degradative recursive fine-tuning on the Qwen backbone. The following sections examine whether this transition generalizes across architectures and update mechanisms.

4.2. Cross-backbone thresholds

To test whether the regime transition observed on Qwen is architecture-specific, we evaluated corresponding rank configurations on two additional backbones: Gemma 3 1B, with five seeds at the boundary ranks over ten generations, and Gemma 4 E2B, with a single seed across three ranks as additional validation.

4.2.1. Gemma 3: cross-backbone replication

On Gemma 3, an analogous regime structure emerges, but at a substantially lower effective rank. Preliminary single-seed experiments identified the regime boundary between $r = 4$ (effective rank i3) and $r = 16$ (effective rank i9); higher ranks were not evaluated with multiple seeds because $r = 16$ already produces clear progressive degradation, making additional high-rank conditions redundant for establishing the boundary location. Table 4 summarizes the results. At $r = 4$ (mean effective rank 3.07), Gemma 3 remains stable across five seeds through ten generations (mean Gen10 retention 94.4%, range 93.5–95.7%). At $r = 16$ (mean effective rank 9.2), Gemma 3 enters the degradative regime across all five seeds (mean Gen10 retention 56.5%, range 52.2–60.9%). The ranges do not overlap at any generation from Gen1 onward, providing consistent empirical separation between the two regimes on this backbone.

Critically, the degradation at $r = 16$ does not stabilize between Gen5 and Gen10. Mean retention declines from 69.1% at Gen5 to 56.5% at Gen10, a further loss of 12.6 percentage points with no sign of plateauing. This progressive trajectory contrasts with the bounded behavior observed at $r = 128$ on Qwen, where retention stabilizes near 89%. The Gemma 3 $r = 16$ configuration is therefore classified as progressively degradative rather than bounded.

A notable feature of the degradative regime on Gemma 3 is that the mean effective rank remains constant at approximately 9.2 throughout all ten generations, even as retention declines from 85% to 57%. This suggests that effective rank operates as a regime gate—determining whether the system enters the degradative regime—rather than as a continuous predictor of the rate of degradation within that regime.

To localize the threshold more precisely, we additionally evaluated intermediate ranks $r \in \{10, 12, 14\}$ with a single seed. Intermediate ranks were evaluated only to localize the transition boundary and were therefore not extended to Gen10. All three exhibited progressive degradation at Gen5: retention of 78.3% ($r = 10$, effective rank 6.37), 73.9% ($r = 12$, effective rank 7.35), and 69.6% ($r = 14$, effective rank 8.37). These values place the Gemma 3 regime boundary between effective ranks 3 and 6.

Table 4

QLoRA rank sweep on Gemma 3 1B IT. Boundary ranks ($r = 4$ and $r = 16$) are evaluated with five seeds over ten generations; intermediate ranks ($r = 10, 12, 14$) with a single seed at Gen5.

Rank	Gen	Retention	Eff. Rank	Regime
$r = 4$	10	94.4% (N=5)	3.07	Homeostatic
$r = 10$	5	78.3%	6.37	Degradative
$r = 12$	5	73.9%	7.35	Degradative
$r = 14$	5	69.6%	8.37	Degradative
$r = 16$	10	56.5% (N=5)	9.17	Degradative

4.2.2. Gemma 4 E2B: additional validation

To provide additional validation from a third model family, we evaluated Gemma 4 E2B across three ranks ($r = 4$, $r = 16$, $r = 64$) with a single seed over ten generations. Table 5 summarizes the results.

The same monotonic dose-response structure emerges: $r = 4$ (effective rank 2.04) remains homeostatic at 94.7% through Gen10; $r = 16$ (effective rank 5.1) exhibits bounded behavior at 88.2%; and $r = 64$ (effective rank 13.3) enters the degradative regime, declining to 75.0% by Gen10. The $r = 64$ configuration additionally exhibits the same distributional drift pattern observed in the Qwen degradative regime: mean response length inflates from 3.6 to 4.4 words, Distinct-1 collapses from 0.55 to 0.43, and stopword ratio increases from 0.23 to 0.28.

Because these results are based on a single seed, we treat them as additional qualitative validation rather than as a precise threshold estimate. The key observation is that a third backbone, from a different model family, exhibits the same regime structure under increasing adapter rank.

Table 5

QLoRA rank sweep on Gemma 4 E2B IT (seed 15, Gen10). Single-seed results provide additional validation of the dose-response structure.

Rank	Retention	Eff. Rank	K_0	Regime
$r = 4$	94.7%	2.04	76	Homeostatic
$r = 16$	88.2%	5.10	76	Bounded
$r = 64$	75.0%	13.28	76	Degradative

4.2.3. Cross-backbone comparison

Figure 4 presents the retention trajectories for all three backbones, enabling direct visual comparison of the regime structure across architectures.

The homeostatic-to-degradative regime transition is observed consistently across all three evaluated backbones, with consistent structure: bounded or stable retention below the threshold, progressive degradation above. However, the threshold location differs substantially across architectures. The Gemma 3 boundary lies between effective ranks 3 and 6, roughly an order of magnitude below the Qwen threshold (between effective ranks 50 and 88). The E2B threshold falls between effective ranks 5 and 13, intermediate between the other two backbones. This ordering does not correspond simply to model size (Gemma 3 1B < Qwen 1.5B < E2B 2B), indicating that the threshold is influenced by architectural or pretraining properties not captured by parameter count alone.

An attempt was made to normalize effective-rank thresholds across architectures by dividing by hidden dimension, number of layers, number of attention heads, and baseline diversity. None of these normalizations reduced the discrepancy to a common threshold. We report this as a negative result and discuss it further in Section 6.

Having established that the regime structure generalizes across architectures, with the key visual features visible in Figure 5: (i) all three backbones exhibit a transition from blue (homeostatic) to red (degradative) markers as effective rank increases; (ii) the transition occurs at substantially different x-axis positions across backbones; and (iii) the Qwen curve remains in the homeostatic region at effective ranks where Gemma 3 is already fully degradative. We next examine whether the same transition occurs when perturbation magnitude is varied independently of adapter capacity.

4.3. Perturbation magnitude: FFT comparison

The previous sections established that increasing update capacity through adapter rank shifts the system from homeostatic to degradative regimes. This section asks whether increasing perturbation magnitude

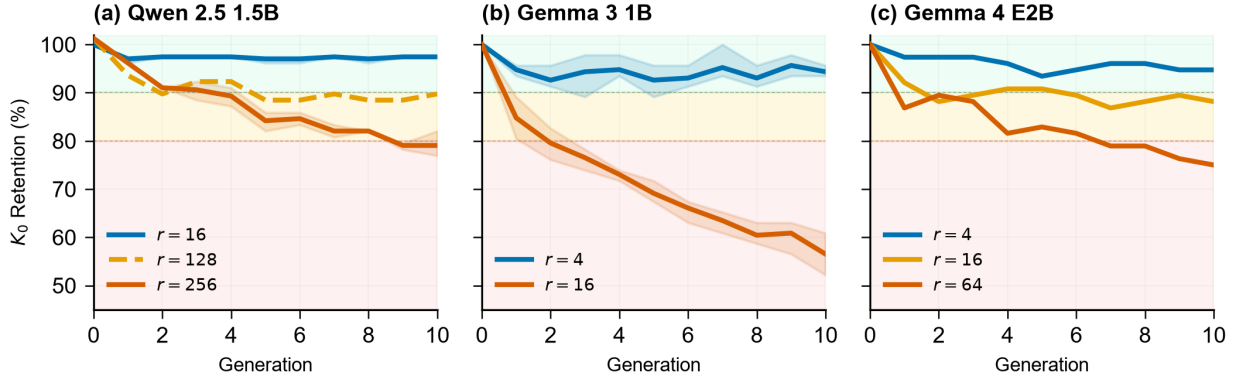


Figure 4: K_0 retention trajectories over ten recursive generations across three backbones: (a) Qwen 2.5 1.5B, (b) Gemma 3 1B, and (c) Gemma 4 E2B. Shaded bands show the range across seeds ($N=5$ for Gemma 3, $N=3$ for Qwen); solid lines show means. All three backbones exhibit consistent regime structure (homeostatic below threshold, degradative above), with the transition occurring at backbone-dependent effective-rank values.

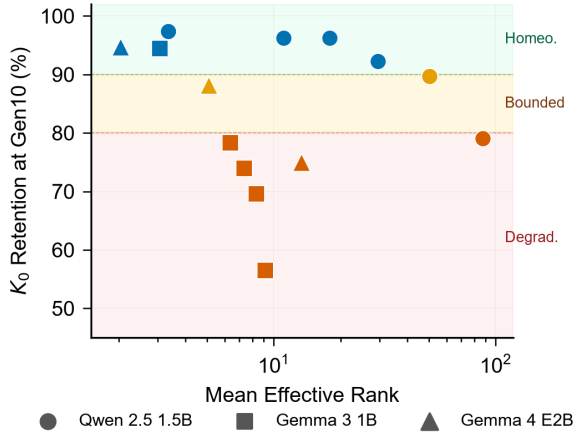


Figure 5: K_0 retention versus mean effective rank across three backbones. Marker shapes distinguish backbones (circle: Qwen, square: Gemma 3, triangle: E2B); colors indicate regime membership (blue: homeostatic, orange: bounded, red: degradative). The regime transition occurs at substantially different effective-rank values across architectures.

through full fine-tuning, without varying adapter rank, reproduces an analogous dose-response pattern.

The learning rate was swept across four values on Qwen 2.5 1.5B under the same recursive replace-only protocol. Table 6 shows retention at Gen3, the earliest generation at which consistent regime separation emerges across all four learning rates. Retention decreases monotonically as learning rate and weight drift increase. Drift values span nearly an order of magnitude (0.39 to 3.48), and retention decreases from 92.3% to 84.6% across this range. These results indicate that perturbation magnitude alone can drive the system toward degradation without any adapter or rank constraint.

4.3.1. QLoRA versus FFT at matched perturbation

To compare the two update mechanisms at similar perturbation, we selected FFT at $LR = 10^{-6}$ (drift 0.39) as the closest available match to QLoRA $r = 16$

Table 6

FFT learning-rate sweep on Qwen 2.5 1.5B (seed 15, Gen3). Perturbation proxy is the mean absolute weight drift relative to the original model for FFT, and BA_F for QLoRA. These are operationally comparable but not geometrically identical measures. QLoRA $r = 16$ is included as a reference point.

Method	LR	Pert. proxy	Gen3 Retention
QLoRA $r = 16$	10^{-5}	0.42 (BA_F)	96.2%
FFT	10^{-6}	0.39	92.3%
FFT	5×10^{-6}	1.56	91.0%
FFT	10^{-5}	1.95	89.7%
FFT	2×10^{-5}	3.48	84.6%

($BA_F \approx 0.42$). Both were run for 10 generations with three seeds. As illustrated in Figure 6, the separation between the trajectories is established after the first recursive generation and remains nearly constant thereafter.

Both methods are stable from Gen1 through Gen10, with no progressive degradation within the observed horizon. QLoRA retains a consistent advantage of approximately 5.6 percentage points (mean Gen10: QLoRA 97.4% vs FFT 91.9%). This gap emerges entirely at Gen1 and does not widen over subsequent generations. This result is therefore interpreted as a difference in one-time adaptation cost rather than a differential degradation rate: FFT incurs a larger initial factual cost when adapting to the synthetic stream, while both methods exhibit similarly stable trajectories once adapted.

QLoRA Gen10 retention is 97.4% across all three seeds; FFT achieves 91.9% (range 91.0–92.3%). The ranges do not overlap across any seed pairing.

Item-level analysis reveals an asymmetry between the two methods. Under FFT, the same six K_0 items

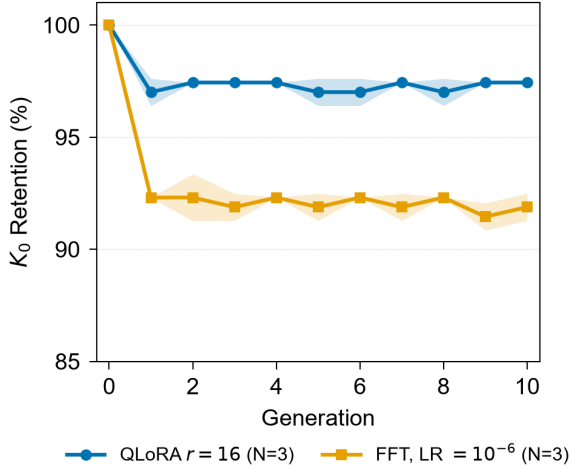


Figure 6: K_0 retention over 10 recursive generations for QLoRA $r=16$ and FFT at $LR=10^{-6}$ on Qwen 2.5 1.5B. Lines show the mean across three seeds; shaded bands indicate ± 1 standard deviation.

are lost across all three seeds (cross-seed Jaccard similarity 1.0), indicating a deterministic fragile-fact floor: these facts are consistently displaced by full-parameter updates regardless of seed. Under QLoRA, only two to three facts are lost per seed, and the overlap between facts lost by QLoRA and facts lost by FFT is zero across all seed pairs (Jaccard 0.0). In this low-perturbation regime, low-rank adaptation loses fewer facts and loses different facts than full fine-tuning at comparable perturbation magnitude.

This pattern is referred to as a stable fragile-fact floor under FFT: certain facts are reliably displaced by full-parameter updates, whereas low-rank updates preserve those same facts while occasionally losing others. This observation is consistent with prior work suggesting that LoRA and full fine-tuning can produce qualitatively different update patterns even when achieving similar downstream performance [44, 15]. The precise mechanistic basis of this divergence in the recursive setting remains an open question for future work.

4.3.2. Rank and learning rate interact

The preceding analyses vary adapter rank and learning rate independently. To test whether their effects are additive or interactive, we evaluated a 3×3 matrix on Qwen: ranks 16, 64, and 256 crossed with learning rates 5×10^{-6} , 10^{-5} , and 2×10^{-5} . Table 7 reports Gen5 retention for each cell, chosen as the common evaluation horizon adopted for the Gemma 3 experiments, enabling cross-backbone comparison.

The interaction is evident (Figure 7): configurations that remain stable under one learning rate become bounded or degradative under another. At $r=16$, only $LR=2 \times 10^{-5}$ pushes the system below 95% retention. At $r=64$, the same degradation appears already at $LR=10^{-5}$. At $r=256$, even the lowest evaluated

Table 7

Rank $\times$ learning rate interaction matrix on Qwen 2.5 1.5B (retention evaluated at Gen5; $K_0 = 78$ items defined at generation 0, seed 15).

Rank	$LR=5 \times 10^{-6}$	$LR=10^{-5}$	$LR=2 \times 10^{-5}$
$r=16$	76/78 (97.4%)	76/78 (97.4%)	72/78 (92.3%)
$r=64$	76/78 (97.4%)	72/78 (92.3%)	69/78 (88.5%)
$r=256$	70/78 (89.7%)	66/78 (84.6%)	64/78 (82.1%)

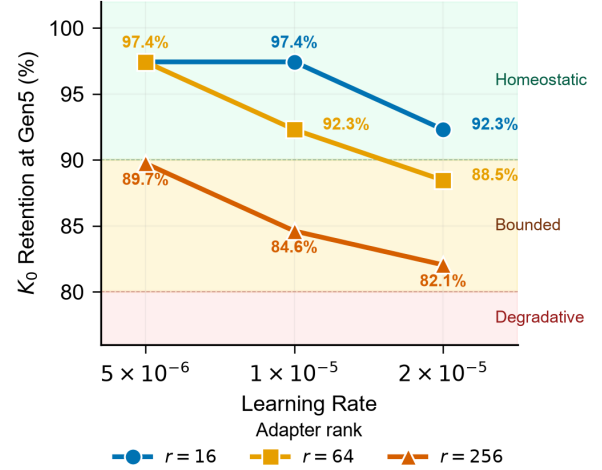


Figure 7: Rank $\times$ learning-rate interaction on Qwen 2.5 1.5B (Gen5 retention). Non-parallel slopes indicate that rank and learning rate jointly determine the operating regime: higher rank amplifies the effect of increased learning rate, and vice versa.

learning rate (5×10^{-6}) produces measurable degradation (89.7%). Conversely, reducing learning rate partially offsets the effect of higher rank: $r=256$ with $LR=5 \times 10^{-6}$ retains 89.7%, compared to 84.6% at $LR=10^{-5}$ and 82.1% at $LR=2 \times 10^{-5}$. This demonstrates that the observed regime depends jointly on update capacity and perturbation magnitude rather than on either axis in isolation.

4.3.3. FFT on Gemma 3 confirms cross-backbone magnitude effect

To test whether the perturbation-magnitude axis generalizes beyond Qwen, we evaluated full fine-tuning on Gemma 3 1B at three learning rates. Gen5 retention was 78.3% at $LR=10^{-6}$, 54.3% at $LR=5 \times 10^{-6}$, and 63.0% at $LR=10^{-5}$. Although the two higher learning rates do not follow a strictly monotonic pattern (possibly reflecting single-seed variability), substantially higher degradation is observed at larger learning rates compared to $LR=10^{-6}$. This confirms that the perturbation-magnitude axis operates on Gemma 3, producing substantially more degradation under higher learning rates and establishing that both the capacity

and magnitude axes generalize across the evaluated architectures.

Both update capacity (Axis 1) and perturbation magnitude (Axis 2) each produce dose-response relationships with factual retention, and these axes interact rather than operating independently. The regime transition therefore reflects a multidimensional pressure landscape in which multiple control variables jointly determine recursive stability. The next section examines how these regimes manifest in the distributional properties of the synthetic outputs.

4.4. Distributional signatures of degradation

The previous sections identified regime transitions through factual retention alone. This section asks how these regimes manifest in the distributional properties of the synthetic outputs. We apply the output-distribution diagnostics defined in Section 3.6 to three representative Qwen configurations selected from the rank sweep: $r = 16$, which exhibited homeostatic behavior; $r = 128$, which exhibited bounded retention; and $r = 256$, which exhibited progressive degradation.

4.4.1. Homeostatic regime ($r = 16$)

In the homeostatic regime, all distributional metrics remain stable across ten generations. Mean response length increases only slightly (2.47 to 2.70 words), Distinct-1 remains above 0.63, MTLTD stays above 2000, stopword ratio remains near 0.16, and baseline response persistence holds steady at approximately 36%. Content efficiency declines modestly from 0.41 to 0.35, reflecting the initial one-generation adaptation cost but no progressive drift.

4.4.2. Bounded regime ($r = 128$)

The $r = 128$ configuration reveals a dissociation between factual retention and output-distribution quality. While K0 retention remains bounded at 89.7% through Gen10, the output distribution degrades substantially: mean response length inflates from 2.47 to 7.0 words, content efficiency collapses from 0.41 to 0.13 (a threefold dilution), stopwords double from 12% to 25%, MTLTD collapses from 2673 to 745, and baseline persistence falls to 5.2%. We refer to this as a *filler verbosity* phenotype: the model pads responses with repetitive function words while retaining enough factual content to maintain bounded retention.

This dissociation has a practical implication: K0 retention alone does not fully capture the onset of distributional degradation. A system monitored solely by factual accuracy would classify $r = 128$ as stable, while its generated outputs have already degraded substantially in efficiency, diversity, and response anchoring.

4.4.3. Degradative regime ($r = 256$)

At $r = 256$, nearly all metrics diverge from their baseline values. Mean response length increases from 2.47 to 3.6 words, content efficiency declines from 0.41 to 0.22, and baseline response persistence collapses from 33% to 3.8% by Gen10. Notably, MTLTD *increases* (2673 to 3804) and stopword ratio *decreases* (12% to 9%), indicating that the model generates more varied vocabulary while losing factual anchoring. We refer to this as an *elaborative drift* phenotype: the model produces longer, lexically diverse, but factually unanchored responses.

The two above-threshold configurations therefore exhibit distinct distributional failure modes despite both departing from stable behavior: $r = 128$ degrades through filler accumulation, while $r = 256$ degrades through elaborative dispersion.

These observations motivate a descriptive three-regime taxonomy:

1. **Homeostatic** ($r \leq 64$ on Qwen, retention above approximately 90%): factual retention and output-distribution metrics are both stable.
2. **Distributionally degraded, factually bounded** ($r = 128$ on Qwen): factual retention remains bounded while output-distribution metrics degrade substantially.
3. **Factually degradative** ($r = 256$ on Qwen): factual retention declines progressively and output distribution drifts.

This taxonomy summarizes the observed behavior of the evaluated configurations rather than defining universally distinct categories. The degradative regime ($r = 256$) is qualitatively consistent with the Stage B “knowledge collapse” described by Keisha et al. [7], where fluency persists while factual accuracy deteriorates. The distributionally degraded regime ($r = 128$), however, represents a condition not captured by that framework: factual retention remains bounded while output-distribution quality degrades substantially. Figure 8 illustrates the divergent trajectories across these three regimes.

The figure reveals three distinct patterns. For $r = 16$ (homeostatic), all three metrics remain flat, confirming that the output stream is stable. For $r = 256$ (degradative), all metrics diverge from baseline simultaneously with retention loss. The most informative case is $r = 128$: response length doubles and content efficiency collapses to one-third of its initial value, yet factual retention remains above 89%, demonstrating that distributional degradation can proceed substantially before factual accuracy signals any problem.

To examine whether output drift precedes factual loss, we computed cross-lag correlations between mean response length at generation t and retention at generation $t+1$. The raw Pearson correlation in the degradative regime is strong ($\rho = *0.965$), but robustness

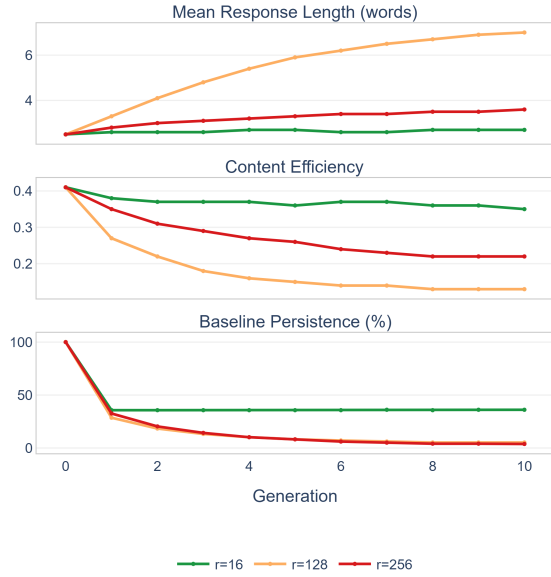


Figure 8: Distributional diagnostics over ten recursive generations for three representative Qwen configurations. (a) Mean response length, (b) baseline response persistence, (c) content efficiency. All three metrics diverge for $r = 128$ and $r = 256$ while remaining stable for $r = 16$; the $r = 128$ configuration is notable for strong distributional drift despite bounded factual retention.

analyses substantially attenuate this association. After conditioning on shared temporal trend (partial correlation), the coefficient drops to $*0.22$. First-difference analysis yields $*0.43$, and the reverse-lag correlation is of comparable magnitude, indicating a bidirectional association rather than clear temporal precedence.

Output drift is therefore interpreted as a co-symptom of the same underlying high-pressure process, rather than a demonstrated causal antecedent of factual loss. Output drift remains valuable as a diagnostic signature: it is consistently present in degradative and distributionally degraded configurations, consistently absent in stable configurations, and can appear even when factual retention remains bounded, as illustrated by the $r = 128$ configuration. Determining the causal relationship between output-distribution changes and factual degradation remains an open question.

The next section tests whether the third component of effective training pressure, synthetic data exposure, can shift the system back toward homeostasis at the regime boundary.

4.5. Synthetic exposure interventions

The preceding sections established that update capacity and perturbation magnitude each govern the regime transition. This section tests the third component of effective training pressure: whether reducing synthetic data exposure can shift the system back from a degradative to a near-homeostatic configuration. All interventions are conducted on Qwen $r = 256$,

which sits at the boundary between the bounded and degradative regimes.

Table 8 summarizes K0 retention at Gen5 for each intervention strategy relative to the unmodified C1 baseline.

Table 8

Synthetic-exposure interventions on Qwen $r = 256$. C3: three seeds; C5: three independent masks. Means reported for multi-seed/mask conditions.

Condition	Ret. (Gen5)	Δ	Tok.	N
C1 Normal	83.3% (65/78)		57k	1
C2 Short	91.0% (71/78)	+7.7 pp	60k	1
C3 Filtered	92.3% (mean)	+9.0 pp	54k	3
C4 Canonical	88.5% (69/78)	+5.1 pp	52k	1
C5 Random	92.7% (mean)	+9.4 pp	54k	3

All four interventions improve retention relative to the C1 baseline. The strongest effects are observed for C3 (length-filtered, +9.0 pp) and C5 (C3-matched random downsampling, +9.4 pp), both of which shift the system from the degradative range (83.3%) to near-homeostatic levels (92–93%).

The critical comparison is between C3 and C5. C3 removes examples with long responses (those exceeding 5 words), while C5 removes a random subset of examples to approximately match C3’s token budget. Despite removing largely disjoint sets of examples (Jaccard overlap between removed sets: 0.015), both produce nearly identical improvements. C3 achieves mean Gen5 retention of 92.3% (range 91.0–93.6% across three seeds); C5 achieves 92.7% (range 92.3–93.6% across three independent masks). The consistency across replicates confirms that the stabilization effect does not depend on which specific examples are removed.

These results suggest that in the Qwen $r = 256$ boundary configuration, the system is sensitive to marginal changes in synthetic exposure volume rather than to the particular quality characteristics of the removed examples. Removing approximately 5% of the training stream under either evaluated selection strategy shifts the configuration toward near-homeostatic retention.

C2 (short-answer constrained) provides an important additional control. It retains all 2,000 training examples and uses slightly more total tokens than C1 (160k vs 157k), yet still improves retention by 7.7 percentage points. This is inconsistent with a simple token-budget explanation in which fewer training tokens mechanically reduce overfitting: the improvement persists even when total token count increases. The token difference between C1 and C3/C5 is only approximately 5% (57k vs 54k), further indicating that the

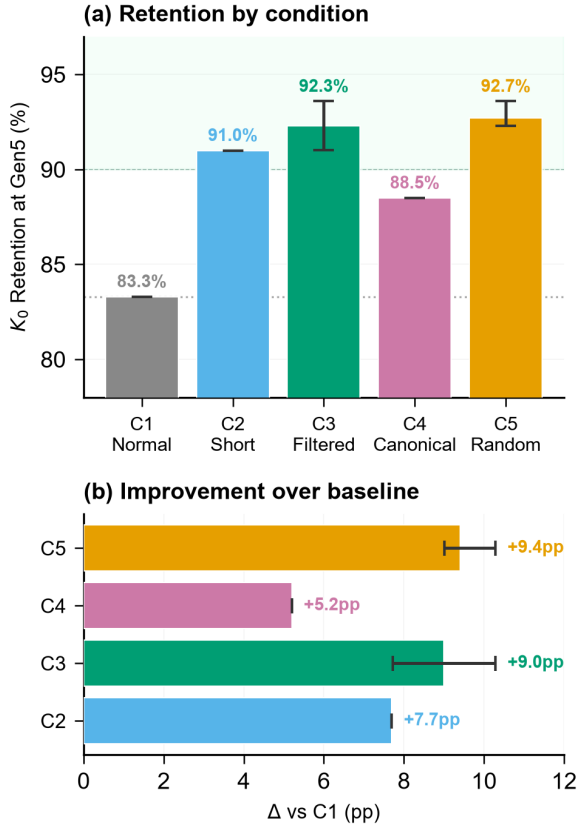


Figure 9: Intervention comparison on Qwen $r = 256$ at Gen5. (a) K_0 retention by condition (C1–C5). (b) Retention improvement relative to C1 baseline. Error bars for C3 and C5 reflect the range across three seeds or masks.

effect operates near an abrupt boundary rather than through gradual token reduction. Token count alone therefore cannot explain the observed recovery.

At the item level, the C1 baseline loses about eight K_0 facts between Gen0 and Gen5. C3 loses only one additional fact beyond the initial Gen1 adaptation; C5 shows a similar pattern. The interventions therefore do not merely delay degradation; they substantially reduce the number of facts displaced during the recursive process.

Figure 9 visualizes the retention and effect-size comparison across conditions.

These results demonstrate that the Qwen $r = 256$ configuration lies near an abrupt regime boundary: a reduction of approximately 5% in synthetic exposure volume is sufficient to restore near-homeostatic retention in this boundary case. This sensitivity is tested only at the $r = 256$ boundary on Qwen and should not be interpreted as a universal property of recursive fine-tuning. Whether comparable boundary sensitivity exists at other ranks, on other backbones, or with different datasets remains untested.

Taken together, the intervention experiments show that the degradative regime is reversible under the evaluated conditions. Adjusting synthetic exposure restores

near-homeostatic behavior in the boundary configuration, complementing the capacity and perturbation results presented in the preceding sections. We note that verification-based approaches [11] represent an orthogonal mitigation axis (filtering by quality rather than quantity); our exposure reduction experiments test boundary sensitivity rather than proposing a competing mitigation pipeline.

Collectively, the five experimental axes establish that recursive stability under PEFT is governed by a multidimensional pressure landscape: update capacity, perturbation magnitude, and synthetic exposure each independently shift the system between regimes, these axes interact rather than operate in isolation, and the regime boundary is backbone-dependent. The following section interprets these findings in the context of prior work.

5. Discussion

The preceding results identify consistent patterns in how recursive synthetic fine-tuning responds to changes in update capacity, perturbation magnitude, and synthetic exposure. This section interprets these findings in the context of prior work, examines what they imply for the understanding and management of recursive training systems, and discusses the limitations of the present study. The discussion is organized around four themes: the interpretation of effective training pressure as an organizing framework (Section 5.1), the relationship between the present findings and the model-collapse literature (Section 5.2), the diagnostic role of output-distribution changes (Section 5.3), and the practical implications for the design and monitoring of recursive training pipelines (Section 5.4).

5.1. Effective training pressure as framework

The results presented in Section 4 converge on a consistent empirical pattern: recursive factual degradation is not observed uniformly across recursive synthetic fine-tuning, but instead emerges only when the training process operates above a backbone-dependent pressure threshold. Three independently manipulated variables, adapter rank, learning rate, and synthetic exposure, each shift the system between homeostatic and degradative behavior. The convergence of these independent manipulations is the primary empirical motivation for treating them as manifestations of a common organizing principle.

The framework is empirical rather than theoretical. We do not propose a closed-form expression that combines rank, learning rate, and exposure into a single scalar predictor. Instead, the experiments reveal that each variable independently controls the same qualitative transition, and that the transition is sharp: approximately 5% reduction in synthetic exposure at the $r = 256$ boundary configuration is sufficient to shift

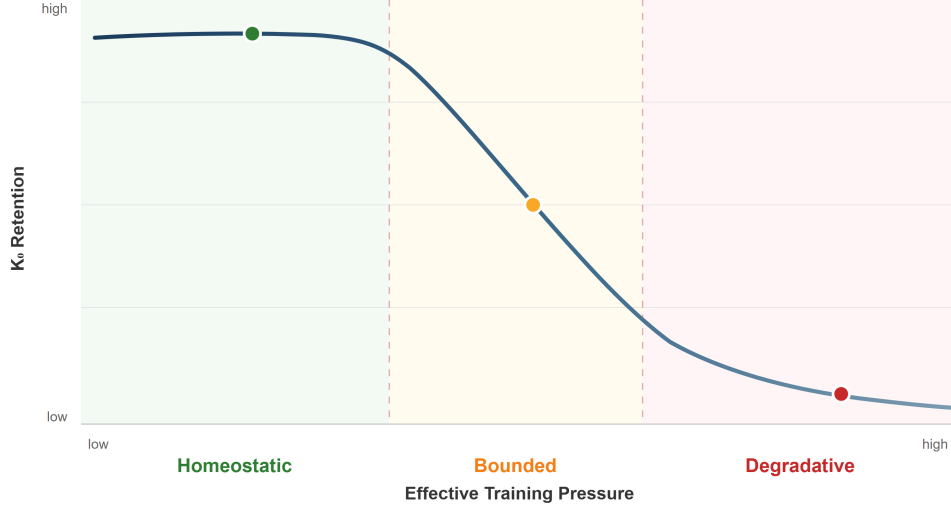


Figure 10: Pressure-response relationship guiding recursive fine-tuning stability. Factual retention transitions through three regimes as effective training pressure increases. Dashed lines mark backbone-dependent boundaries; the transition is abrupt and its location varies by roughly an order of magnitude across architectures.

the system from degradative to bounded behavior. This sharpness suggests that the underlying phenomenon operates near an abrupt regime boundary rather than degrading gradually as any single parameter increases.

The contribution is the existence and reproducibility of this sharp transition, not the numerical threshold itself. The present work does not claim that the specific effective-rank values at which regimes change will generalize to arbitrary models or tasks. What appears to generalize is the qualitative structure: a pressure-dependent boundary separating stable from degradative recursive behavior, controllable through multiple axes.

The rank $\hat{r}$ learning-rate interaction experiment (Table 7) further demonstrates that these axes are not independent: different combinations of rank and learning rate produce similar recursive outcomes, indicating that the regime depends jointly on both variables. This result strengthens the interpretation of effective training pressure as a multidimensional control landscape rather than a collection of separable single-variable effects. Beyond the interaction itself, two additional convergences support this interpretation: the transition boundary induced by each axis shares the same qualitative signature (abrupt rather than gradual), and the same distributional drift patterns accompany degradation regardless of which axis crosses the threshold. These convergences would be coincidental under a model of three independent failure modes, but are naturally explained by a shared underlying sensitivity to the magnitude of per-generation parameter perturbation. Developing a quantitative formulation that combines update capacity, perturbation magnitude, and exposure into a single predictive scalar remains an important direction for future work.

Table 9

Evidence matrix: mapping framework components to supporting experiments.

Observation	Supporting evidence
Rank governs transition	Rank sweep, 6 values, $N=3$ at $r=16$, $r=256$
Magnitude governs transition	FFT LR sweep, 4 values
Exposure governs transition	Intervention C3/C5, $N=3$ masks
Threshold is backbone-dependent	Qwen vs Gemma 3 (10 $\hat{r}$ gap)
Drift accompanies degradation	5 distribution metrics
Transition is sharp	5% exposure shift flips regime
Capacity $\hat{r}$ magnitude interact	Rank $\hat{r}$ LR matrix (3 $\hat{r}$ 3)
Generalizes cross-backbone	Gemma 3 ($N=5$) + E2B dose-response

Table 9 summarizes how each component of the framework is supported by independent experimental evidence.

Two features of this framework merit emphasis. First, the threshold is backbone-dependent: Gemma 3 transitions at effective rank 3–6, while Qwen remains outside the degradative regime up to effective rank 150. No simple normalization by architectural parameters (hidden dimension, depth, head count) aligned these thresholds. This likely reflects properties of the pre-trained backbone not captured by simple architectural dimensions. Second, the framework does not assume that the three pressure components are interchangeable or commensurable. Rank and learning rate operate through different mechanisms (subspace dimensionality versus step size), and synthetic exposure operates on

the data stream rather than the update itself. Their shared effect on the regime transition is the empirical observation that motivates the framework, not a theoretical prediction.

5.2. Relation to recursive-training literature

The model-collapse literature has established that recursive training on synthetic data can produce irreversible degradation under full fine-tuning [6] and that expected loss increases linearly with any nonzero fraction of synthetic data [8]. Our results do not contradict these findings. Rather, they help localize the experimental conditions under which qualitatively similar degradation emerges in our recursive fine-tuning setting: the degradative configurations ($r = 256$ on Qwen, $r \geq 16$ on Gemma 3, FFT at high learning rates) are qualitatively consistent with the collapse dynamics these works describe.

The contribution of the present work is locating where, in the space of practical PEFT configurations, the transition between bounded and degradative behavior occurs. Biderman et al. [15] observed that LoRA “learns less and forgets less” relative to full fine-tuning. Empirical studies of catastrophic forgetting in LLMs during continual fine-tuning [45] have similarly observed scale-dependent forgetting patterns. Our dose-response results refine this observation: low-rank adaptation remains bounded not because LoRA prevents degradation in general, but because, in the evaluated configurations, low rank limits effective training pressure below the observed regime boundary. When rank is increased sufficiently ($r = 256$ on Qwen, $r = 16$ on Gemma 3), degradation emerges despite using the same PEFT method. These findings therefore provide an empirical boundary for the qualitative observation reported by Biderman et al. [15], showing not only that low-rank adaptation forgets less, but also where this advantage ceases to hold.

The cross-backbone results add a dimension absent from prior work. Both Shumailov et al. [6] and Dohmatob et al. [8] treat collapse as a property of the recursive process itself, without examining its dependence on model architecture. Our finding that the threshold differs by approximately an order of magnitude between Qwen and Gemma 3 indicates that the pretrained backbone contributes substantially to determining where the boundary lies. This is consistent with the observation by Keisha et al. [7] that Gemma 3 1B exhibits three-stage degradation under full fine-tuning, a finding compatible with our result that Gemma 3 has a lower pressure threshold than Qwen. However, because Keisha et al. [7] used a different dataset, evaluation format, and training protocol, we treat this as suggestive context rather than a controlled comparison.

Table 10

Scope of generalization: what the present evidence supports versus what remains unknown.

Supported by evidence	Remains unknown
Regime transition exists	Threshold at larger scales
Pressure dependence	Exact functional form
Sharp (not gradual) boundary	Behavior beyond Gen10
Reproducibility (3–5 seeds)	Transfer to non-factoid tasks
Cross-backbone regime structure	Cross-architecture normalization
Capacity & magnitude interaction	Exposure sensitivity cross-backbone
Magnitude axis cross-backbone	Interaction strength at scale

The synthetic-exposure interventions connect to the data-axis mitigation literature. Gerstgrasser et al. [10] showed that accumulating real data across generations prevents collapse, operating on the data-composition axis. Wang et al. [46] further showed that synthetic data diversity significantly impacts downstream model quality. Our C3/C5 results operate on a complementary axis: rather than adding real data, we reduce synthetic exposure by a marginal amount and observe recovery in a boundary configuration. Both approaches reduce recursive degradation through different intervention points in the training pipeline. They are orthogonal and potentially combinable, though their interaction has not been tested.

The present work complements rather than challenges the collapse literature by identifying the practical conditions under which bounded and degradative recursive behavior are observed in parameter-efficient fine-tuning, how the operating regime varies across backbones, and how it can be manipulated through multiple independent control axes. In this sense, the present study offers a practical response to the theoretical pessimism of prior work: recursive degradation, while real, is not inevitable under the configurations most commonly used in practice. This aligns with the recent observation by Schaeffer et al. [47] that the term “model collapse” encompasses multiple distinct phenomena, and that empirical regime characterization may be more productive than treating collapse as a monolithic threat.

To clarify the scope of the present findings, Table 10 distinguishes what the evidence supports from what remains open.

5.3. Distributional degradation as a diagnostic signal

The distributional analysis in Section 4.4 reveals that factual retention alone provides an incomplete

picture of recursive degradation. The $r = 128$ configuration on Qwen retains 89.7% of K_0 facts through Gen10, a level that could reasonably be interpreted as stable under a retention-only monitoring strategy. Yet its output distribution degrades substantially: content efficiency collapses to 0.13, baseline persistence falls to 5.2%, and MTLTD drops from 2673 to 745. This dissociation between factual retention and distributional quality provides the principal motivation for using output-distribution diagnostics as complementary monitoring signals.

Two distinct degradation phenotypes emerge above the pressure threshold. At intermediate capacity ($r = 128$), the model accumulates filler words and repetitive function tokens while preserving enough factual content to maintain bounded retention. At high capacity ($r = 256$), the model instead generates lexically diverse but factually unanchored responses. These patterns appear to represent distinct modes of distributional deterioration rather than different degrees of the same phenomenon: they differ in which metrics degrade (MTLTD collapses in the first case, increases in the second) and in their relationship to factual loss (bounded in the first, progressive in the second).

The elaborative drift phenotype is not unique to Qwen: the Gemma 4 E2B $r = 64$ configuration exhibits the same pattern (response length inflation from 3.6 to 4.4 words, Distinct-1 collapse from 0.55 to 0.43, stopword ratio increase from 0.23 to 0.28), suggesting that this distributional signature generalizes across backbone families when effective training pressure exceeds the regime threshold.

The practical implication is that monitoring recursive training systems solely through factual accuracy or loss metrics risks missing the onset of distributional drift. Content efficiency, baseline persistence, and response length provide richer signals of regime departure. In the $r = 128$ case, these metrics showed substantial degradation across generations during which retention remained bounded, suggesting that they could serve as earlier indicators of regime departure than retention alone.

This diagnostic value does not rest on a causal claim. The cross-lag analysis showed that after controlling for shared temporal trend, the association between output drift and subsequent retention loss weakens substantially (partial $\rho = *0.22$). The analysis suggests that output drift is a co-symptom of the same high-pressure process rather than a demonstrated causal driver. Its value lies in being observable, measurable, and consistently present in above-threshold and distributionally degraded configurations, while remaining low in stable configurations, regardless of whether it contributes to, or simply accompanies, factual degradation.

5.4. Practical implications

The results and interpretations above translate into actionable considerations for practitioners designing or monitoring recursive synthetic fine-tuning pipelines. The following discussion is organized around three operational concerns: configuration selection, runtime monitoring, and intervention design.

Regarding configuration selection, the dose-response results indicate that adapter rank is a directly controllable parameter with predictable effects on recursive stability. For applications where recursive synthetic data reuse is anticipated, selecting a rank that places the system well below the observed pressure threshold is a practical strategy for maintaining recursive stability. On Qwen 2.5 1.5B, ranks at or below 64 (effective rank ≤ 30) remained stable through ten recursive generations with no sign of drift. Although the numerical threshold is backbone-dependent, the underlying principle appears consistent across the evaluated models: lower effective training pressure is associated with greater recursive stability in the tested configurations. When full fine-tuning is required, lower learning rates reduce recursive degradation in the evaluated setting. Notably, this form of control requires no external verifier, no additional real data, and no modification to the training objective; it operates entirely through configuration choices already available to practitioners.

Regarding runtime monitoring, the $r = 128$ dissociation demonstrates that factual accuracy alone is insufficient for detecting distributional drift. Based on the observed patterns, tracking at minimum mean response length, content efficiency (retention normalized by length), and baseline response persistence can help detect regime departure earlier than factual accuracy alone. Rising response length or declining content efficiency signals that the system may be departing from its stable operating regime, even if factual retention metrics appear stable. These signals are computationally inexpensive and require no additional evaluation infrastructure beyond what is already needed for generation-level quality assessment.

Regarding intervention design, the C3/C5 results demonstrate that boundary configurations are sensitive to marginal changes in synthetic exposure. When monitoring signals indicate regime departure, reducing the volume of the synthetic training stream through filtering, downsampling, or constraining response format may provide a lightweight mitigation strategy under conditions similar to those evaluated here. Under the evaluated conditions, random downsampling performed comparably to targeted filtering, suggesting that identifying specific problematic examples may not always be necessary.

In summary, practitioners deploying recursive synthetic fine-tuning should consider:

- Keeping adapter rank or learning rate below the empirically characterized pressure threshold for the target backbone.
- Monitoring distributional diagnostics (output length, content efficiency, response persistence) alongside factual accuracy.
- Reducing synthetic exposure when distributional drift is detected, even before retention metrics decline.
- Calibrating pressure thresholds independently for each model architecture, as cross-backbone transfer of numerical values is unreliable.

An exploratory replication of the synthetic-exposure intervention on the Gemma 3 boundary configuration ($r=16$, using the same approximate 5% random down-sampling protocol) did not reproduce the stabilization observed on Qwen. Under the evaluated conditions, Gemma 3 retention followed the same trajectory with and without the intervention. While update-capacity and perturbation-magnitude effects generalized across the evaluated backbones, exposure sensitivity at the tested magnitude remains demonstrated only for the Qwen boundary configuration. This asymmetry reinforces that effective training pressure should not be interpreted as implying identical sensitivity along all axes across architectures.

These considerations are derived from experiments on 1–2B parameter models with factoid QA data. Their applicability to larger models, different domains, or longer recursive horizons remains to be established. They should therefore be viewed as empirically grounded design considerations rather than universal prescriptive rules.

6. Limitations

The interpretation of these findings is subject to several limitations.

All experiments use models in the 1–2B parameter range (Qwen 2.5 1.5B, Gemma 3 1B, Gemma 4 E2B). Whether these findings generalize to 7B, 70B, or larger models has not been established in this work. Larger models may exhibit different pressure thresholds for reasons not evaluated in the present study. This scale is consistent with prior model-collapse studies [6, 7] and reflects the computational constraints of running ten recursive generations with multiple seeds and configurations.

The evaluation uses a single dataset (TriviaQA) comprising short factoid question-answer pairs. TriviaQA was chosen specifically to isolate factual retention from reasoning or stylistic confounds: short answers minimize format-dependent evaluation noise, and the substring-matching criterion is resilient to changes in verbosity. The findings may not extend to long-form

Table 11

Cross-architecture threshold normalization attempts. Threshold r values: Qwen 150–88, Gemma 3 13–6, E2B 15–13. None of the tested normalizations aligned thresholds across backbones.

Norm.	Qwen	Gemma 3	E2B	Aligned?
r / d	0.033–0.057	0.003–0.005	0.003–0.008	No
r / $\sqrt{d}$	1.28–2.24	0.09–0.18	0.13–0.33	No
r / layers	1.79–3.14	0.12–0.23	0.14–0.37	No
r / heads	4.17–7.33	0.75–1.50	0.63–1.63	No
r / KV heads	25.0–44.0	3.0–6.0	5.0–13.0	No

generation, open-ended dialogue, code synthesis, or other domains where output structure and evaluation criteria differ substantially.

The recursive horizon extends to ten generations. The long-term dynamics beyond Gen10 are unknown: configurations classified as stable over this horizon may eventually degrade given sufficient generations, and configurations classified as bounded may stabilize or worsen.

Multi-seed evaluation is used for headline comparisons (three seeds), while supporting ablations are evaluated with a single seed. This provides consistent descriptive patterns but limited statistical power for formal inference. The observed consistency across seeds supports consistency under the evaluated conditions but does not substitute for larger-scale replication.

Cross-architecture threshold normalization was attempted using several architectural ratios, none of which aligned the thresholds between Qwen and Gemma 3 (Table 11).

The raw effective-rank threshold is therefore not portable across backbones without empirical calibration, limiting its predictive utility for new architectures.

The perturbation-matching comparison between QLoRA and FFT relies on approximate proxies (BA_F for QLoRA versus mean absolute weight drift for FFT). These quantities are operationally comparable but not geometrically identical, introducing residual confounding in the method comparison. Additionally, all QLoRA experiments use 4-bit NormalFloat quantization of the base model. Although the quantized base remains static across generations and adapters are trained in higher precision, the interaction between quantization error and high-rank adapter updates has not been isolated. A comparison against standard 16-bit LoRA would be needed to fully disentangle quantization effects from rank effects.

The synthetic-exposure interventions (C3/C5) are tested only at the $r = 256$ boundary on Qwen. Whether comparable sensitivity exists at other ranks, on other backbones, or with different datasets has not been established. The restriction to a single boundary configuration is deliberate: the purpose of these experiments

is to demonstrate pressure sensitivity at the regime transition, not to propose a general mitigation method. Whether the same intervention would recover stability under extreme pressure (e.g., FFT at high learning rate) remains untested.

All mechanistic analysis operates at the output level (retention counts, response properties, cross-lag correlations). No weight-level analysis of adapter update subspaces, gradient geometry, or representational drift was conducted. The causal mechanisms by which effective training pressure produces regime transitions remain outside the scope of the present study.

The output-distribution diagnostics (content efficiency, persistence, SDI-3) are evaluated only under the present experimental conditions. Their effectiveness as monitoring signals in production settings, including reliability across domains and deployment conditions, has not been assessed. Regarding the K_0 metric itself, substring matching could in principle confound verbosity-induced format changes with genuine knowledge loss. However, the C2 intervention (which constrains response format to short answers) improves retention without fully restoring baseline levels, indicating that the degradation measured by K_0 reflects actual factual displacement rather than format masking alone.

Finally, the present work establishes which variables shift which regimes, but does not yet derive a quantitative formulation combining the three pressure components into a single scalar predictor. Whether such a formulation can be obtained, for instance through multivariate regression on a larger configuration space, remains an open question for future work.

7. Conclusion

Through systematic dose-response experiments across three backbones, ten recursive generations, and multiple seeds, we identified effective training pressure as an organizing framework for recursive stability under parameter-efficient fine-tuning. The central finding is that recursive degradation is a pressure-gated phenomenon: increasing update capacity, perturbation magnitude, or synthetic exposure each independently shifts the system from homeostatic to degradative behavior, with the transition being abrupt, backbone-dependent, and reproducible. Cross-backbone replication further shows that degradation in the above-threshold regime is progressive and does not stabilize within the observed horizon. At the regime boundary, marginal reductions in synthetic exposure restore near-homeostatic retention, although this sensitivity is backbone-dependent. Distributional diagnostics further reveal that output quality can degrade substantially before factual loss becomes apparent.

The practical implication is that recursive degradation need not be an inevitable consequence of training on synthetic data. The boundary between stability and

degradation can be controlled through configuration choices already available to practitioners, provided that the relevant backbone-dependent pressure threshold has been identified. The contribution is the identification of this reproducible regime structure, not the specific numerical thresholds, which are expected to vary across architectures, scales, and tasks.

Future work should scale these experiments to larger models (7B+), extend the evaluation to diverse output domains, and develop a quantitative formulation that unifies the observed pressure components into a predictive model. Weight-level analysis of adapter update geometry could illuminate the mechanistic basis of the observed transition, and combining multiple pressure-reduction strategies within a single pipeline could reveal whether their effects are additive or interactive.

A. Project repository and data access

All source code, configuration files, and experimental outputs developed for this paper are publicly available at *[Repository URL removed for blind review]*. These resources ensure full reproducibility of the experimental protocol and allow other researchers to replicate or extend the reported results.

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